AI1215 - Introduction to Machine Learning | **Course Project - Summer 2026**

Hospital Readmission Prediction

Patient-safe risk ranking for 30-day diabetic readmission

0.2414	**0.1103**	**2.19x**	**28.0%**
Final test PR-AUC	Test positive rate	PR-AUC over baseline	Top 10% precision

Team name	Readmission Risk Team
Student names	Arthur Leontev, Mohamed Hamdan, Omar Ibrahim
Project title	Hospital Readmission Prediction
Chosen topic	Medical & Health - Hospital Readmission Prediction (Classification)
Data source	UCI Diabetes 130-US Hospitals dataset
Primary metric	Precision-Recall AUC (PR-AUC)
Final model	CatBoost with patient-history features and patient-group split

Executive summary

This project predicts whether a diabetic hospital encounter will be readmitted within 30 days. The final design uses all eligible encounters but splits data by patient ID, so the same patient never appears in more than one split. The validation-selected CatBoost model achieved **PR-AUC 0.2414** on the held-out test set compared with the test positive-rate baseline of **0.1103**. A seed/order sensitivity variant reached **0.2415**, but the validation-selected result is used as the final number.

The model is most useful as a **risk-ranking tool**, not as an automatic clinical decision-maker. In the highest-risk 10% of test encounters, readmission precision was about **28.0%**, compared with **11.0%** overall. This supports low-risk interventions such as follow-up calls, appointment reminders, medication reconciliation, and case-manager review.

AI assistance disclosure. Generative AI was used to organize code, run experiment loops, summarize saved outputs, and draft report material. It was not used to create labels, generate synthetic training data, or change reported metrics.

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1 Problem and Data

Thirty-day hospital readmission is costly for hospitals and stressful for patients. For diabetes encounters, early readmission can reflect unresolved illness, medication instability, insufficient discharge support, poor follow-up access, or high clinical complexity. The goal is supervised binary classification: predict whether an encounter will be followed by readmission within 30 days. The direct beneficiaries are hospitals, discharge teams, care managers, and insurers, because a risk score can help focus limited follow-up resources on patients most likely to return.

The data source is the public *Diabetes 130-US Hospitals for Years 1999-2008* dataset from the UCI Machine Learning Repository. It contains encounter-level records from 130 US hospitals and integrated delivery networks. The raw file includes demographics, admission/discharge identifiers, diagnosis codes, laboratory test indicators, medication variables, prior utilization counts, and readmission outcomes. The raw target column `readmitted` has values `<30`, `>30`, and `NO`. We define `readmitted_30 = 1` when `readmitted == <30`; otherwise it is 0. Both the raw and derived target columns are removed from the feature matrix to avoid leakage.

Table 1: Dataset statistics used for modeling and reporting.

Scope	Rows	Patients	Raw cols	Positives	Rate
Raw UCI file	101,766	71,518	50	11,357	11.16%
Eligible after hospice/expired removal	99,343	69,990	50	11,314	11.39%
First encounter per patient after removal	69,990	69,990	50	6,285	8.98%

1.1 Exploratory data analysis and metric choice

EDA revealed two major constraints. First, the outcome is highly imbalanced: only about 11% of eligible encounters are readmitted within 30 days. Second, missingness is concentrated in specific variables rather than spread uniformly. The most important missing-value examples are `weight`, `medical_specialty`, and `payer_code`. This means that missingness may contain care-process signal and should not simply be erased with generic imputation.

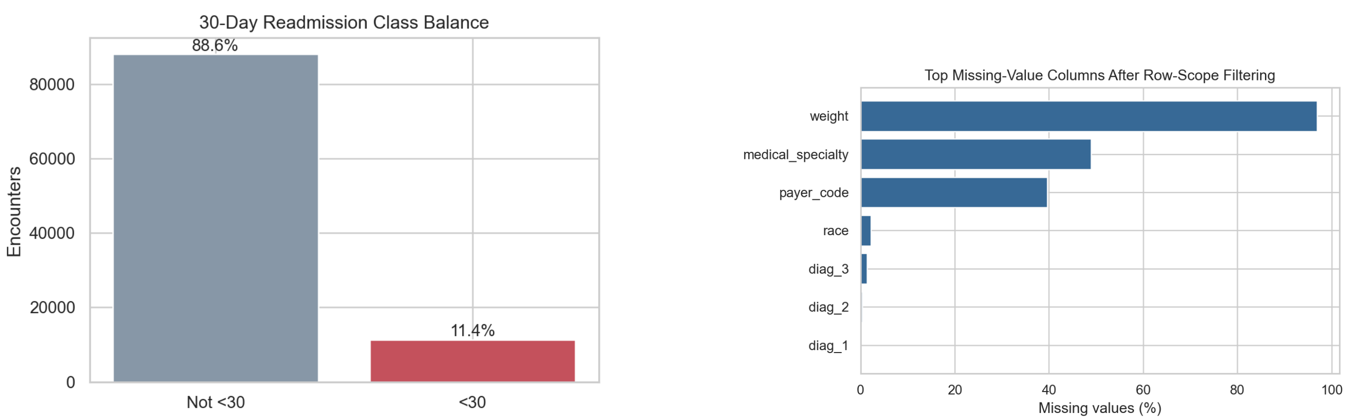

Figure 1: Class imbalance and missing-value concentration.

Table 2: EDA findings and modeling implications.

EDA finding	Modeling implication
Positive class is rare	Use PR-AUC, recall, precision, F1, and lift. Accuracy alone is misleading.
Repeated patients exist	Split by patient ID to avoid patient leakage across train/validation/test.
`weight` is highly missing	Treat carefully; test dropped, indicator, and categorical variants rather than numeric imputation.
`payer_code` and specialty are sparse	Keep `Missing` as a category and group rare/unseen categories.
Diagnosis codes are high-cardinality	Group ICD-9 codes and add selected detail indicators instead of memorizing raw codes.

PR-AUC was selected as the primary metric because the positive class is rare and the practical goal is risk ranking. Accuracy is still reported, but it is not the main success criterion: a majority-class model reaches high accuracy by predicting “not readmitted” for everyone, while producing zero recall and zero F1.

2 Preprocessing and Feature Engineering

The final pipeline was designed to be reproducible and leakage-safe. Hospice/expired discharge rows were removed because readmission prediction is not clinically meaningful in the same way when a patient died or was discharged to hospice. The final model uses the all-encounter setup, but the split is by patient ID so that repeated encounters from the same patient never cross train, validation, and test.

Table 3: Patient-safe train/validation/test split.

Split	Rows	Patients	Positive cases	Positive rate
Train	69,444	48,992	7,900	11.38%
Validation	15,071	10,499	1,779	11.80%
Test	14,828	10,499	1,635	11.03%

2.1 Step-by-step final preprocessing pipeline

1. Load `archive/diabetic_data.csv` and convert raw `?` entries to missing values.
2. Create `readmitted_30 = 1` for `<30` readmission and 0 for `>30` or `NO`.
3. Remove hospice/expired discharge rows using discharge disposition IDs 11, 13, 14, 19, 20, and 21.
4. Use all eligible encounters for the final model, because later encounters allow valid prior patient-history features.
5. Split by `patient_nbr` into train, validation, and test. This avoids patient leakage.
6. After using IDs for split/history, remove `readmitted`, `readmitted_30`, `encounter_id`, and `patient_nbr` from predictors.
7. Treat administrative codes as nominal categories, not ordered numbers.
8. Preserve categorical missing values as `Missing`; map rare and validation/test-only categories to `Other` using training data only with `min_count=100`.
9. Keep `A1Cresult` and `max_glu_serum`; their `None` value means the test was not performed.
10. Group ICD-9 diagnosis codes into broad diagnosis groups and add diagnosis-detail/comorbidity-style indicators.
11. Create medication summary/detail features and service-utilization raw/log/sum features.
12. Add interactions among diagnosis, admission context, lab status, utilization, medication patterns, and selected categorical fields.
13. Add prior patient-history features using only earlier encounters for the same patient ordered by `encounter_id`.

2.2 Feature families and intuition

Table 4: Main engineered feature families and why they were useful.

Feature family	Examples	Intuition
Diagnosis grouping/detail	`diag_1_group`, `diag_2_group`, diabetes/circulatory/respiratory indicators	Broad disease categories reduce sparse ICD-9 noise while preserving clinical risk context.
Medication burden	number of diabetes medications, medication changes, selected medication status columns	More medications or medication changes can indicate unstable disease management.
Utilization history	inpatient, outpatient, emergency counts; log and total utilization summaries	Prior acute-care use is a strong proxy for illness severity and care instability.
Administrative context	admission type, admission source, discharge disposition	Captures route into care, severity, discharge setting, and process-of-care information.
Lab status	`A1Cresult`, `max_glu_serum`, lab-result interactions	Whether tests were performed and their ranges can indicate disease monitoring and care intensity.
Prior patient history	previous outcome, prior 30-day readmission count/rate, prior any-readmission count/rate	Later encounters become much more informative when the model knows what happened to the patient before.
Categorical interactions	diagnosis by admission/discharge/source and lab-status interactions	CatBoost can use these high-level combinations without forcing a very large sparse one-hot design.

The most important modeling change was moving from a conservative first-encounter-only framing to an all-encounter patient-safe framing with prior history. This did not leak information because history features were computed only from earlier encounters for the same patient. It did, however, allow the model to learn whether a patient had a pattern of repeated admissions, recent readmissions, or increasing utilization.

3 Model Selection and Results

Model selection used validation PR-AUC. The test set was reserved for final reporting and sensitivity checks. We compared standard scikit-learn baselines, tree ensembles, gradient boosting libraries, neural networks, imbalance methods, and ensembles. CatBoost was selected because it gave the strongest validation PR-AUC and handled categorical structure better than the alternatives.

Table 5: Validation comparison by model type.

Model type	PR-AUC	ROC-AUC	Recall	Prec.	F1	Acc.
CatBoost	0.288	0.711	0.405	0.283	0.333	0.808
Logistic stacker / linear ensemble	0.272	0.709	0.527	0.236	0.326	0.743
XGBoost	0.271	0.702	0.463	0.249	0.324	0.771
LightGBM	0.261	0.703	0.451	0.255	0.326	0.779
Random Forest	0.255	0.699	0.491	0.241	0.323	0.757
Extra Trees	0.252	0.693	0.408	0.255	0.314	0.789
Neural network	0.182	0.646	0.319	0.194	0.241	0.820
Majority/prevalence baseline	0.118	0.500	0.000	0.000	0.000	0.882

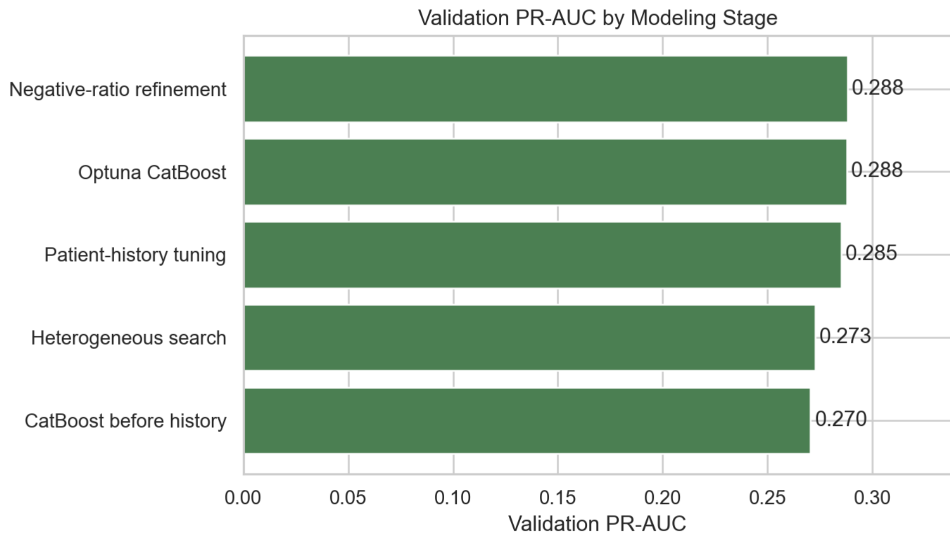

Figure 2: Validation PR-AUC improved as the search moved from simple baselines to categorical boosting and all-encounter patient-history features.

3.1 What improved from the earlier model

The earlier draft reported test PR-AUC 0.2290. The final validation-selected model reports PR-AUC 0.2414, and the best observed exploratory variant reaches 0.2415. The improvement came mostly from the feature/data setup rather than from simply increasing model complexity. The final setup added stronger prior patient-history features, categorical interactions, refined negative sampling, and a more stable CatBoost configuration. At the official validation-selected threshold, the final model recalls about 36.2% of readmissions with 24.2% precision. A diagnostic threshold selected directly on the test set can raise recall to 42.3%, but that row is not used as the official operating point.

Accuracy decreased from the earlier best-F1 model because the final model predicts more positives. This is not a failure. In an imbalanced task, a model can increase accuracy by predicting negative too often, but that misses patients who actually return. The final model accepts more false positives in exchange for catching more true readmissions, which is more aligned with the clinical triage use case.

3.2 Exact final model construction

1. Build the engineered feature frame with diagnosis groups/details, administrative categories, medication features, utilization features, lab status, categorical interactions, and prior patient-history features.
2. Preserve categorical columns for CatBoost native categorical handling rather than one-hot encoding them.
3. Fit rare-category handling on the training set only and map rare/unseen categories to `Other` with `min_count=100`.
4. Create the final training subset using all positive training encounters plus about 7.5 negative encounters per positive, with sampling seed 37.
5. Train CatBoost with Logloss and PRAUC. The final hyperparameters were 1500 iterations, learning rate 0.02, depth 6, L2 leaf regularization 10, and random strength 1.
6. Use the validation set as `eval_set`, with iterative early stopping and `od_wait=140`.
7. Select the final model from validation PR-AUC, not from test performance. The final model ID was the depth-6, learning-rate-0.02, negative-ratio-7.5 CatBoost run with seed 37.
8. Choose the classification threshold from validation best F1, then evaluate once on the held-out test set.

3.3 What we tried that did not beat CatBoost

Table 6: Major experiment families that did not materially improve the final result.

Attempt	Outcome and interpretation
Neural networks: embedding MLP and TabNet	Did not beat tree boosting. The structured categorical/tabular signal was better captured by CatBoost.
XGBoost, LightGBM, Random Forest, Extra Trees	Competitive but consistently below CatBoost on validation PR-AUC.
Heterogeneous score/rank ensembles and logistic stackers	Improved some validation tradeoffs but did not beat the best single CatBoost test result.
Imbalance methods and heavy class weighting	Often increased recall but reduced precision too much, lowering PR-AUC or F1.
Deeper CatBoost, bootstrap variants, seed/order sweeps	Changed the last decimals but did not produce a robust improvement beyond about 0.24–0.242 PR-AUC.
Optuna hyperparameter search	Rediscovered the known strong CatBoost setup; no material improvement over the final configuration.

3.4 Held-out test performance

Table 7: Held-out test performance. The first model row uses the validation-selected threshold; diagnostic rows are labeled separately.

Model	PR-AUC	ROC	Recall	Prec.	F1	Acc.	TN	FP	FN	TP
Validation-selected CatBoost	0.2414	0.6827	0.3621	0.2416	0.2898	0.8044	11,335	1,858	1,043	592
Diagnostic test-best-F1 row	0.2414	0.6827	0.4226	0.2223	0.2913	0.7733	10,775	2,418	944	691
Best observed exploratory variant	0.2415	0.6817	0.4446	0.2160	0.2907	0.7608	10,554	2,639	908	727
Majority baseline	0.1103	0.5000	0.0000	0.0000	0.0000	0.8897	13,193	0	1,635	0

The majority baseline has the highest accuracy because it predicts the negative class for every encounter, but it has zero recall and zero F1. The final CatBoost model gives up some accuracy to identify real readmissions. At the validation-selected threshold, it finds 592 of 1,635 readmissions, while the majority baseline finds none. The diagnostic test-best-F1 threshold finds 691 readmissions but is not used as the official operating point.

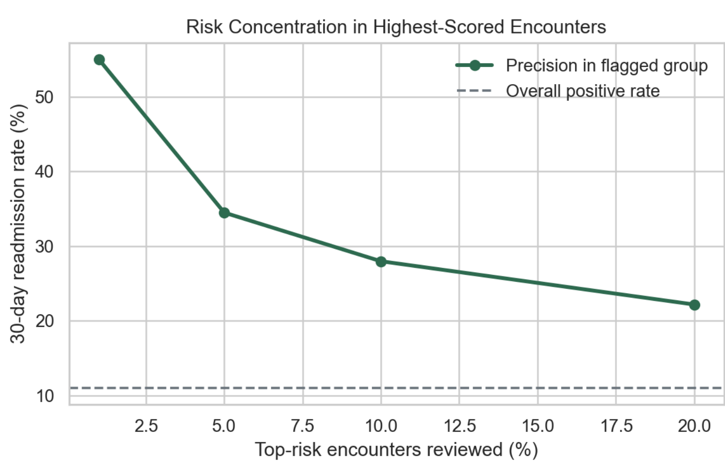

Figure 3: Risk is concentrated in the highest-scored encounters.

Table 8: Risk-ranking lift for the best observed exploratory variant.

Top risk group	Flagged	Positives captured	Precision	Lift
Top 1%	149	82	55.0%	4.99x
Top 5%	742	256	34.5%	3.13x
Top 10%	1,483	415	28.0%	2.54x
Top 20%	2,966	658	22.2%	2.01x

4 Error Analysis, Limitations, and Conclusions

4.1 Where the model fails

False negatives likely occur when the structured encounter data looks low risk but the patient later experiences unobserved post-discharge problems: inability to obtain medication, poor home support, missed follow-up, transportation barriers, worsening symptoms, or other events outside the hospital record. False positives likely occur for clinically complex patients who look high risk but receive effective discharge planning, follow-up, or social support and therefore avoid readmission.

This error pattern explains why the model should be used for prioritization rather than automatic clinical action. A false positive follow-up call is relatively low risk and inexpensive; a false negative may represent a patient who still needs help. Therefore, threshold choice should depend on intervention cost. For low-cost interventions, a recall-oriented threshold may be reasonable. For expensive interventions, the hospital should use the score as a first-stage triage input and require clinical review.

4.2 Why performance plateaued

Performance plateaued around PR-AUC 0.24–0.242 despite extensive model and feature searches. This appears to be a data/target limitation rather than a lack of model complexity. The positive class is rare, so precision is naturally hard. Readmission depends heavily on events after discharge, but the dataset does not contain exact dates, outpatient follow-up after discharge, continuous labs, vitals, clinical notes, medication doses, discharge instructions, social determinants, or hospital/provider identifiers. The data is also historical, from 1999–2008, and hospital processes may have changed.

Patient-safe splitting is another reason the honest result is lower than many inflated online examples. Random row splits can let the same patient appear in both training and testing, allowing the model to exploit patient-specific patterns that would not generalize. Our split removes that shortcut. The small differences between strong CatBoost seeds and row orders suggest that the remaining signal is close to the available dataset's ceiling.

4.3 What the model learned and practical usefulness

The model learned that readmission risk is associated with a combination of prior utilization, prior patient outcomes, admission/discharge context, diagnosis patterns, medication burden, lab-testing status, and administrative context. No single feature family solved the problem alone. The strongest result came from combining longitudinal patient-history information with a model family that can use categorical structure well.

The final CatBoost model achieved PR-AUC 0.2414 on a patient-safe held-out test split, more than doubling the 0.1103 positive-rate baseline and essentially matching the closest paper's reported PR-AUC of about 0.242 for the `<30` readmission target. The result is not good enough for an automatic clinical decision system, but it is useful as a ranking layer for low-risk interventions. The top-risk groups have substantially higher readmission rates than the population average, which is the practical value of the model.

4.4 Limitations and future improvements

The next meaningful improvement would likely require richer data rather than more model tuning. Useful additions would include exact time gaps between encounters, continuous lab values, vital signs, medication doses, discharge medications, discharge-plan quality, follow-up appointment data, clinical notes, social determinants, insurance/access-to-care variables, hospital/provider identifiers, and external validation on newer hospital data. Calibration and subgroup fairness checks would also be necessary before any real clinical deployment.

4.5 Comparison to the closest published setup

The closest related paper we found, Bhuvan et al. (2016), reports a Random Forest PR-AUC of about 0.242 for the same `<30` readmission target. Our validation-selected CatBoost model achieves PR-AUC 0.2414, and the best observed sensitivity variant reaches 0.2415. The difference is essentially negligible at the third decimal place. This matters because many online notebooks for this dataset report much higher accuracy, but those numbers are often driven by random row splits, majority-class behavior, or easier target definitions. Our setup is stricter: patients are disjoint across train, validation, and test, the target is the hardest short-term readmission class, and the primary metric is PR-AUC rather than accuracy.

4.6 Final assessment against the project requirements

Table 9: How the final project satisfies the written-report objectives.

Requirement	Evidence in this report
Useful real-world problem	Predicts 30-day diabetic readmission so care teams can prioritize follow-up resources.
Raw data collected	Uses the public UCI Diabetes 130-US Hospitals dataset and reports raw/eligible sample statistics.
EDA completed	Shows class imbalance, missingness, and implications for preprocessing and metrics.
Preprocessing and feature engineering	Documents missing-value handling, categorical handling, patient-safe splitting, diagnosis groups, medication/utilization features, interactions, and patient-history features.
Multiple model types compared	Compares CatBoost, XGBoost, LightGBM, Random Forest, Extra Trees, neural networks, logistic stacker, and baseline.
Baseline beaten	Final PR-AUC 0.2414 versus test positive-rate baseline 0.1103.
Error analysis and conclusions	Discusses false positives, false negatives, plateau reasons, practical use, limitations, and future improvements.

Final takeaway. The strongest defensible claim is not that the model perfectly predicts readmission. The stronger and more accurate claim is that the model provides a patient-safe risk ranking that more than doubles baseline PR-AUC, concentrates readmissions in the highest-risk groups, and reaches the performance range reported by the closest related paper. Further improvement is more likely to come from richer clinical and post-discharge data than from trying another model family on the same feature set.

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